Non-Convex Mesh Collider. Automatic Generator

v2.0

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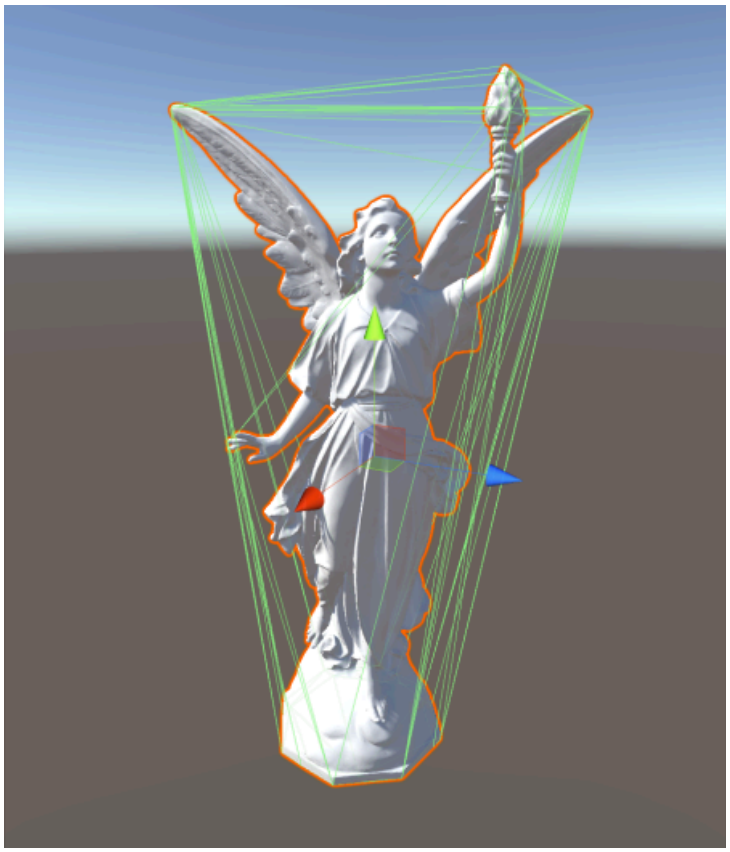
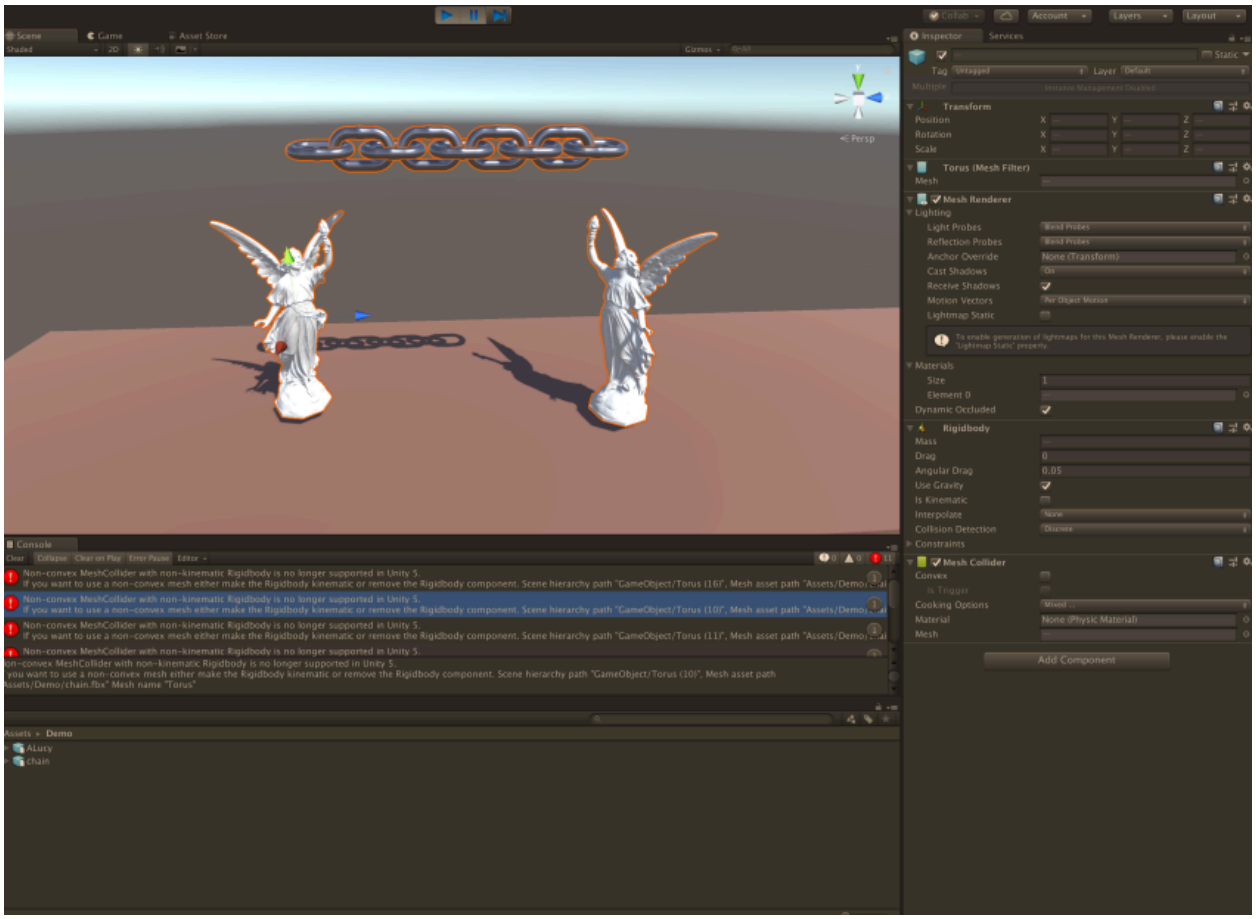
What is the problem? How to solve it?

You want to create something like a scene from this video:

You added a new object to your non-static geometry. And you want to detect collisions and/or do a physics simulation. But what you have is:



This is (obviously) a really bad approximation of your geometry. Ah, that's because it is marked as Convex, so let's disable that…



No, it doesn't work at all: since Unity 5 it is no longer supported! And it's impossible to simulate chain's physics like that at all.

Solution

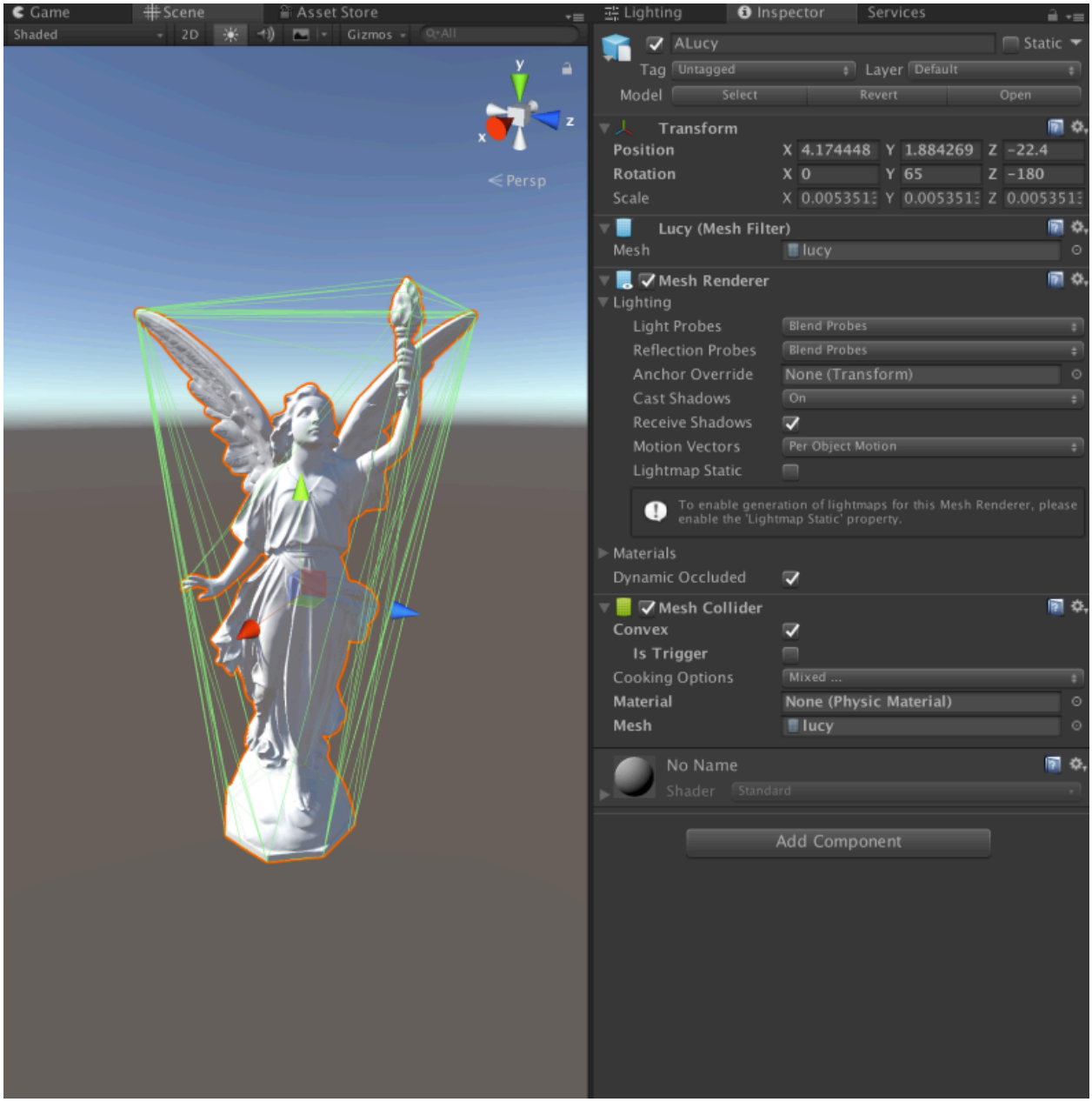
Non-Convex Mesh Collider. Automatic Generator fixes this for you!

It creates a set of convex colliders that together forms the nonconvex collider that you expect. This is industry-standard way of solving this problem and it is used already by other well known engines. It is based on modified V-HACD and COACD libraries that are 3x faster than original ones.

So, how to use it exactly?

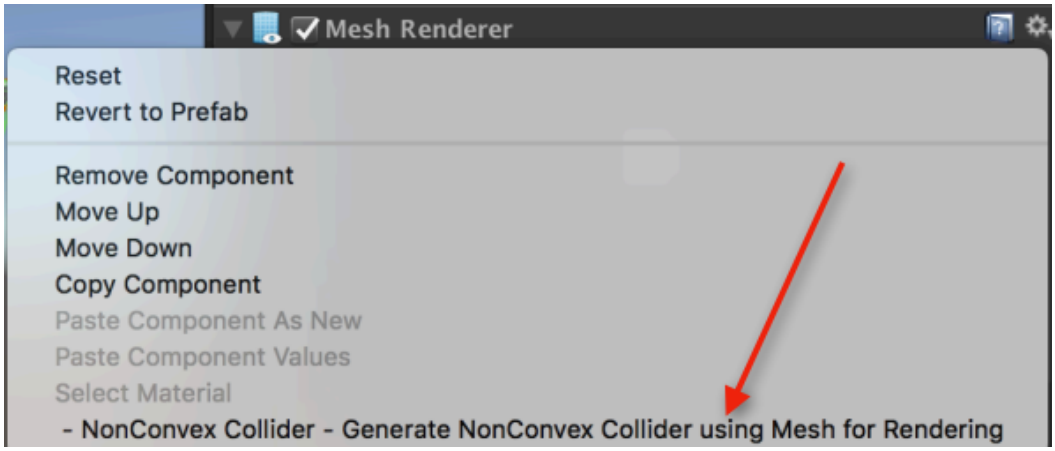
You can use it in the editor to create a Mesh. It is possible to use via Unity Editor Inspector or via UnityEditor compatible code.
Or in Runtime on Desktop platforms only - see RuntimeAPI demo.

Step #1. Select your object

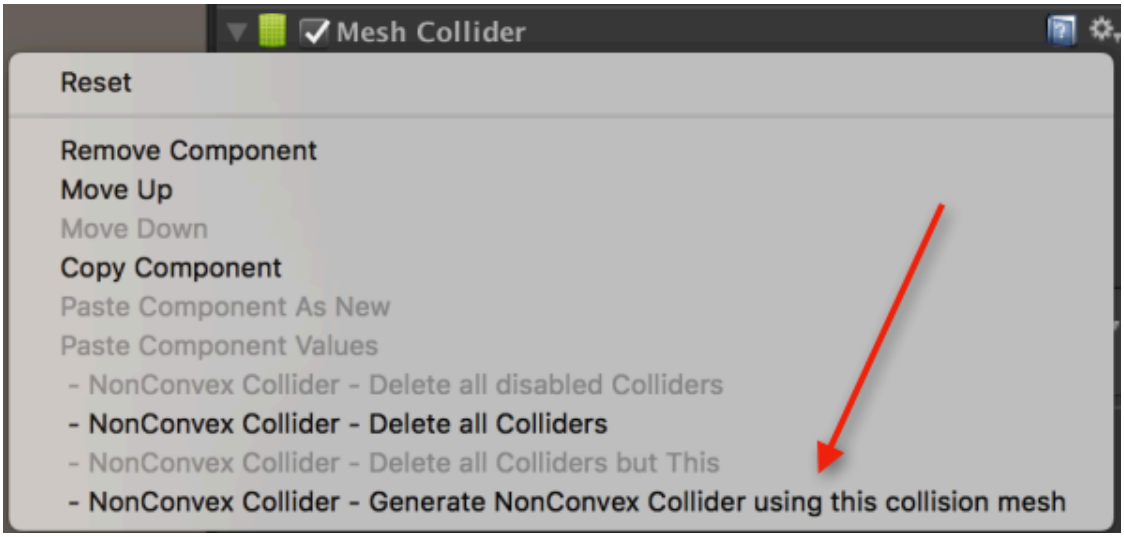


And there are 3 options to create non-convex mesh from this point:

1. Press the gear icon on MeshRenderer or MeshFilter:



2. Press the gear icon on MeshRenderer or MeshFilter:

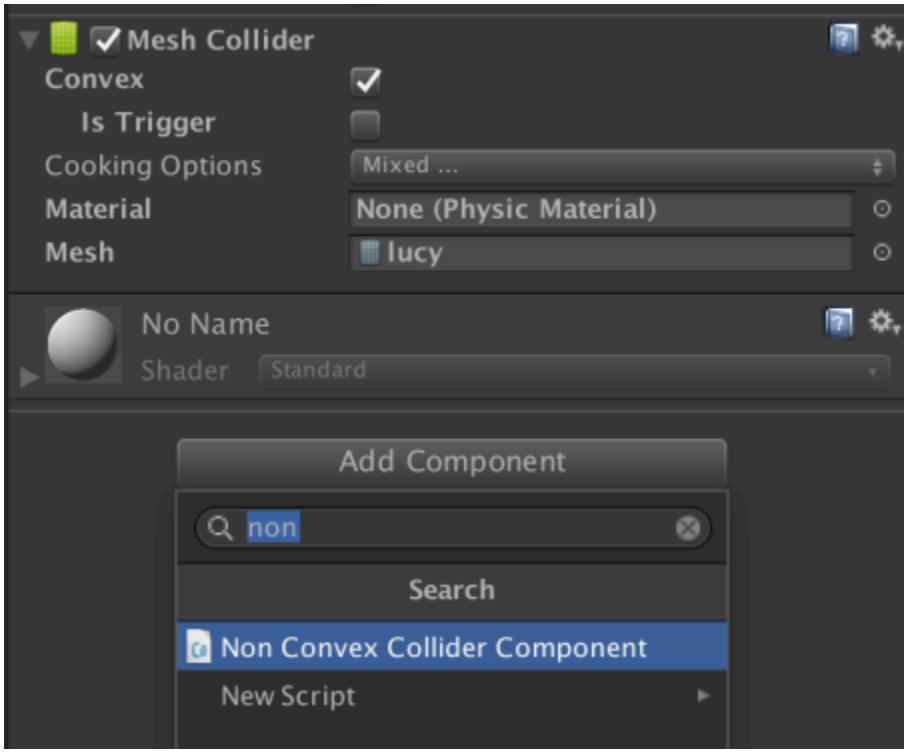


as you can see there are few more options in here.

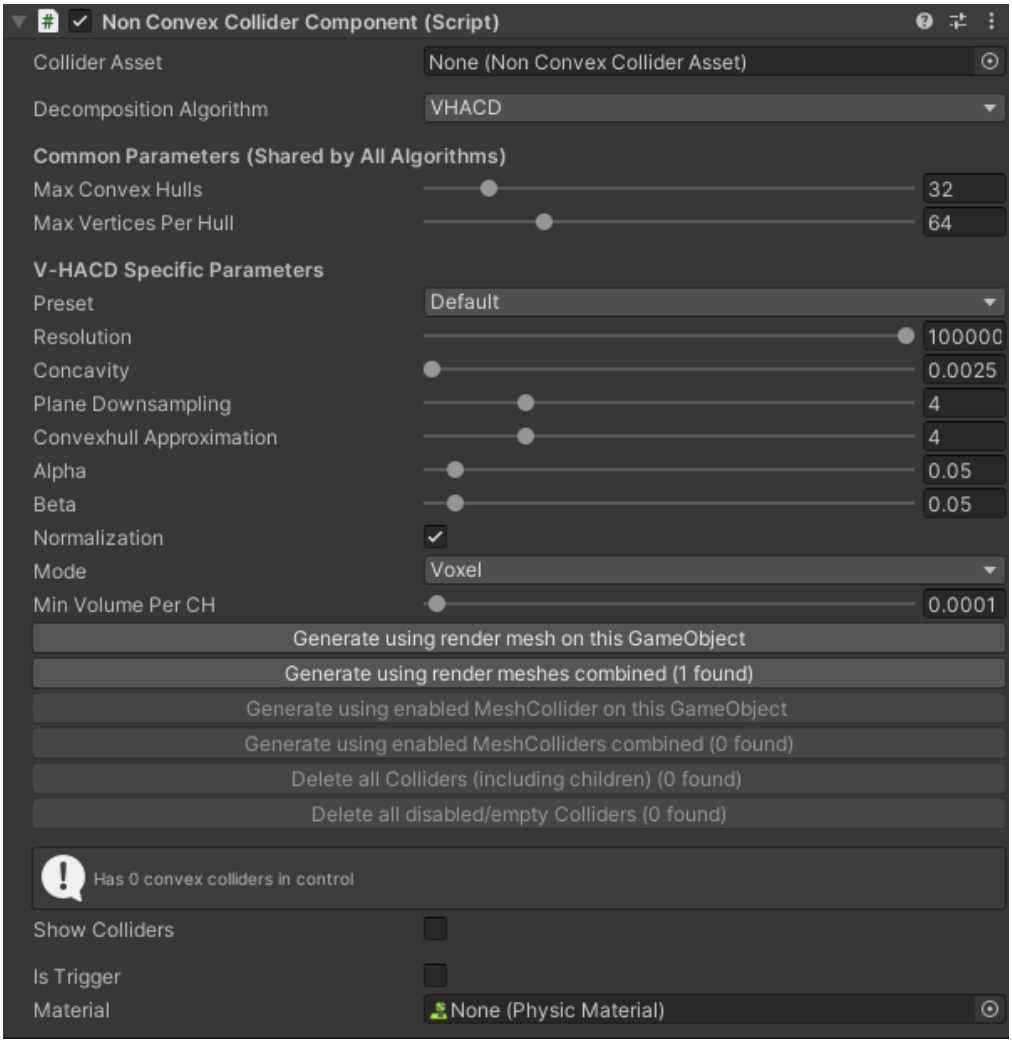
Note:
Delete all disabled Colliders,
Delete all Colliders,
Delete all Colliders but This

will delete the colliders, no matter were they created by this asset or not.

3. Add Non Convex Collider component

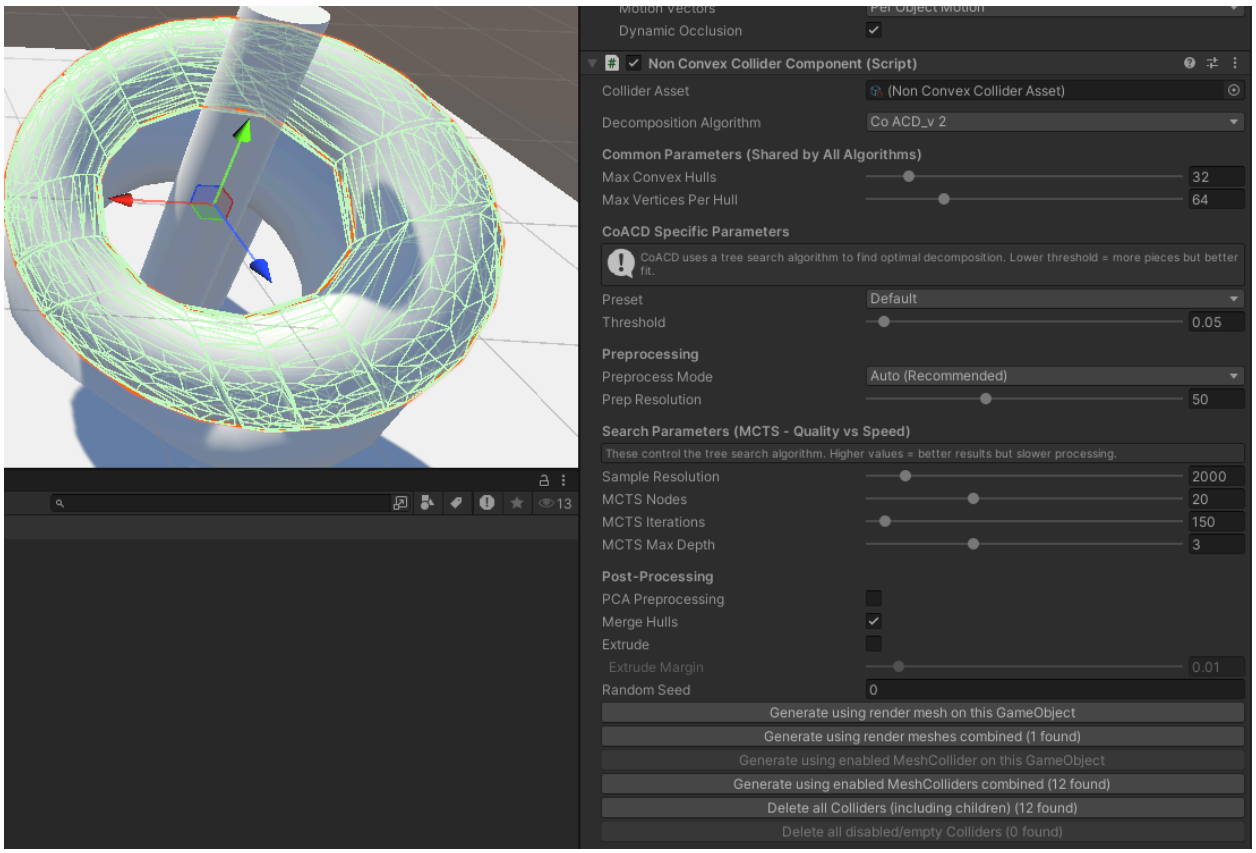


this provides you more options to tweak:



If you click Generate using rendering meshes combined - it will generate the same way as 1) (if you have only one rendering mesh).

If you click Generate using enabled MeshCollider combined - it will generate the same way as 2) (if you have only one enabled rendering mesh). As a result you will have this:



Parameters description is [<here>].

So as you can see - there are lots of convex mesh colliders. You can easily control them using buttons on Non Convex Collider component. Also if you change IsTrigger or Physics Material on it - all colliders will sync up.

Also as you can see - new Collider Asset was generated that contains all convex colliders information inside (you can drag'n'drop already generated ones, for example). Also next time you want to generate - it won't re-generate, but just pick already existing asset with the same settings and meshes.

Runtime Generation

It is possible to generate non-convex colliders in runtime, on your procedurally generated meshes. Runtime Demo is included in the package.

```
using (var _ = new StopwatchScoped("Generate mesh"))
{
  GenerateTorusMesh(mesh);
}

gameObject.AddComponent<MeshRenderer>();

Mesh[] meshes;
using (var _ = new StopwatchScoped("NonConvexCollider generate meshes"))
{
  if (useNewCOACD)
     meshes = API.GenerateConvexMeshesCoACD(mesh, CoACDParameters.Default());
  else
     meshes = API.GenerateConvexMeshes(mesh, Parameters.Default());
}

using (var _ = new StopwatchScoped("NonConvexCollider generate add to gameobject"))
{
  var colliderAsset = NonConvexColliderAsset.CreateAsset(meshes);

  var nonConvex = gameObject.AddComponent<NonConvexColliderComponent>();
  nonConvex.SetPhysicsCollider(colliderAsset);
  nonConvex.Algorithm = useNewCOACD ? DecompositionAlgorithm.CoACD_v2 : DecompositionAlgorithm.VHACD;
}
```

Non-Convex Collider component parameters

Max Convex Hulls

Since physics engine can work only with convex colliders we're just decomposing non-convex colliders to multiple convex colliders.

This setting means - create not more than this number of convex MeshColliders.

The less the faster physics simulation will be, but less precise the result will be.

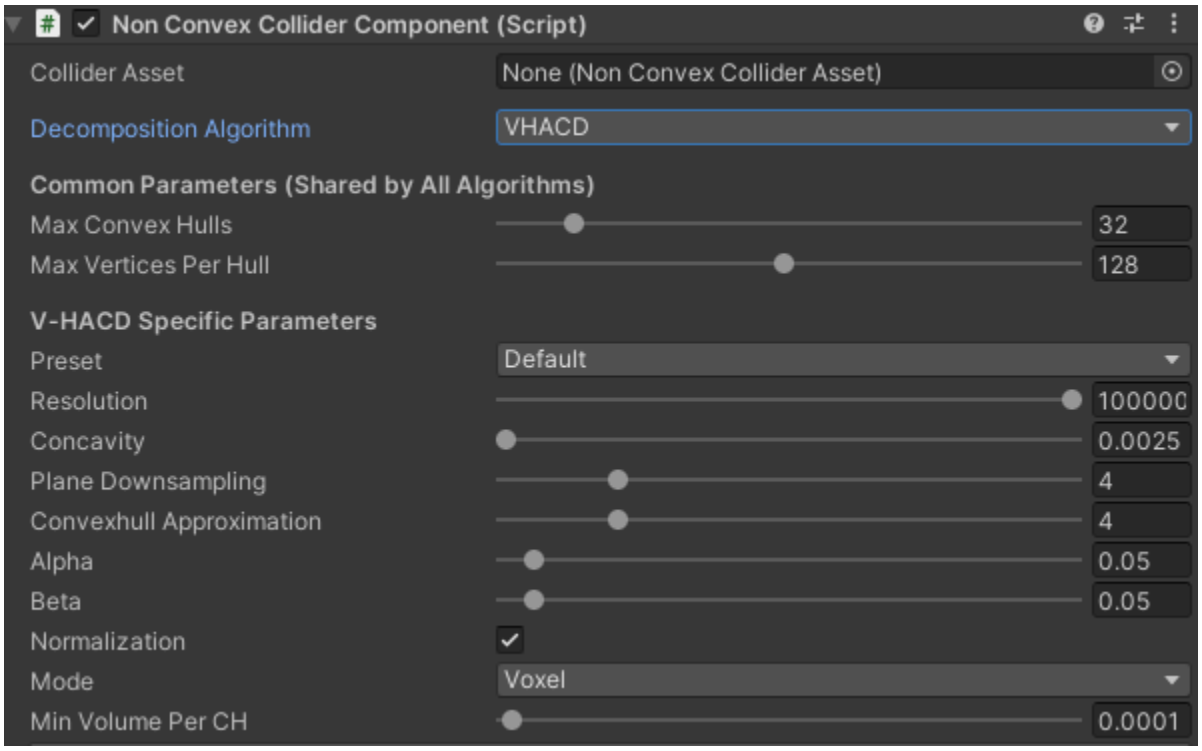
Max Verices per Hull

This controls how complex each of the convex colliders should be. Be aware that Unity has a limitation of 256 triangles per convex collider.

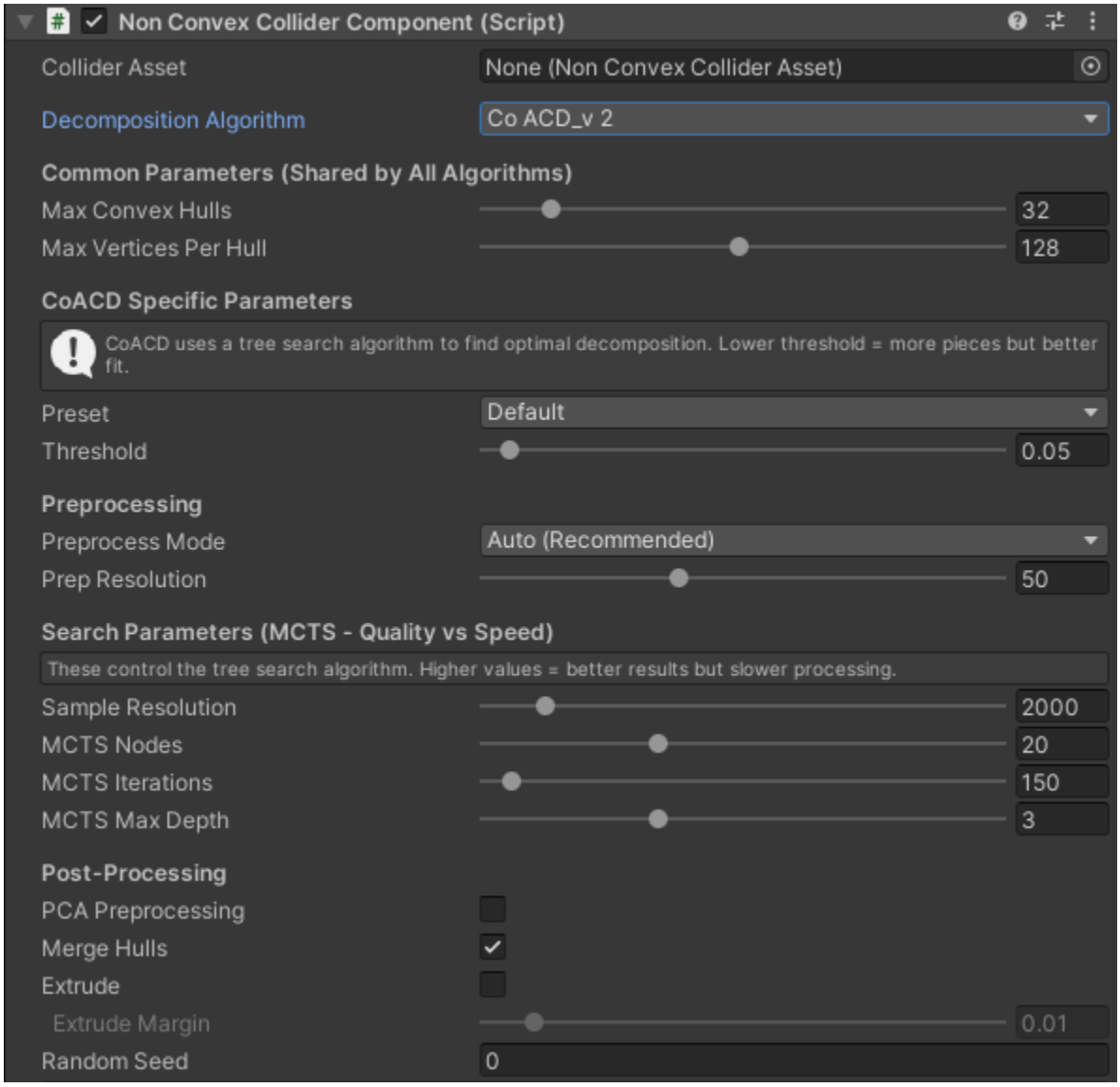
This is again, same thing - the less the faster physics simulation will be, but less precise the result will be.

VHACD

This algorithm is fast, but not as precise.



COACD


Non Convex Collider Component (Script)
Collider Asset
None (Non Convex Collider Asset)
Decomposition Algorithm
Co ACD_v 2
Common Parameters (Shared by All Algorithms)
Max Convex Hulls
32
Max Vertices Per Hull
128
CoACD Specific Parameters
CoACD uses a tree search algorithm to find optimal decomposition. Lower threshold = more pieces but better fit.
Preset
Default
Threshold
0.05
Preprocessing
Preprocess Mode
Auto (Recommended)
Prep Resolution
50
Search Parameters (MCTS - Quality vs Speed)
These control the tree search algorithm. Higher values = better results but slower processing.
Sample Resolution
2000
MCTS Nodes
20
MCTS Iterations
150
MCTS Max Depth
3
Post-Processing
PCA Preprocessing
Merge Hulls
Extrude
Extrude Margin
0.01
Random Seed
0

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V-HACD

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COACD

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